

A uniform entropy bound for fibres of $k\sigma(k)$

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<https://rsihouse.ai/openmath> ; <https://luma.com/yzp9abvr> ;

<https://www.sundai.club/events/boston/recursive-learning-hack-with-harvard-innovation-labs>

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Central source and adjudication archive: <https://github.com/alejandrozu/openmath-2026-judging>

Abstract

The number $f(n)$ of positive integers k satisfying $k\sigma(k)=n$ has an all-target upper bound $\exp((C+\varepsilon)\log n/\log \log n)$, where $C=\log(2539/1000)/3+\log(10/9)/2\approx 0.36327036$. This refines the located $\log 3/3$ coefficient while leaving the requested zero-constant asymptotic open.

The exact arithmetic count

Let $\sigma(k)$ be the sum of positive divisors of k , and let $f(n)=\#\{k\geq 1:k\sigma(k)=n\}$. Since $k\sigma(k)\geq k$ for positive k , the solutions form an explicit finite subset of $\{0,\dots,n\}$; the source proves identity with the canonical formal-conjecture count, including its positivity convention.

For every $\varepsilon>0$, the result gives a threshold N such that all $n\geq N$ satisfy $f(n)\leq \exp((C+\varepsilon)\log n/\log \log n)$, with $C=\log(2539/1000)/3+\log(10/9)/2$. The numerical comparison is all-target: C is smaller than $\log 3/3\approx 0.366204$. A separately located $\log 2/2$ bound for odd targets does not subsume a theorem for all n .

Powerful cores and an exact pattern bound

Split k into its powerful core, containing prime exponents at least two, and a coprime squarefree residual part. The squarefree map $s\mapsto s\sigma(s)$ is injective; the formal argument removes the largest prime and treats the small primes separately. Thus a solution in a fixed target fibre can be encoded by its core exponent pattern without multiplying by an uncontrolled number of squarefree choices.

The equation $k\sigma(k)=n$ constrains both divisibility and size of that core. The development establishes exact weighted exponent-pattern bounds, with local weights and a cubic estimate at a fixed rational parameter. These finite inequalities yield the $2539/1000$ constant, rather than introducing an unproved entropy heuristic.

A logarithmic cutoff divides target primes into small and large ranges. Small-prime choices are bounded crudely but contribute a lower-order term; large-prime exponents obey a log-size budget. The proven weighted finite estimate is then converted to the exponential form, and a real asymptotic bridge absorbs lower-order terms into ε . Every premise of the intermediate conditional bridge is discharged by the final SigmaFiberAsymptotic theorem.

Mathematical and formal boundaries

The requested Erdős 1060 conclusion has a little-o exponent, corresponding to an arbitrarily small leading constant. A fixed positive C does not establish that conclusion. The result is a uniform quantitative partial improvement and should be labelled that way.

Fresh independent replay completed 2026-10-05T01:08:02+00:00: all ten frozen authored Lean modules compiled under Lean 4.33.1 and Mathlib 0df444a360eaa60ab8c11dca51a86af692955474. The actual printed axiom lists for `canonical_count_eventually_upper`, `canonical_count_upper` and `entropyConstant_lt_old` contain only `propext`, `Classical.choice` and `Quot.sound`; all three requested prints are present, with no admitted, native-evaluation or unrecognized axioms. Source hashes, commands, individual outputs and guarded resource measurements are preserved in `Verification/sakana-erdos1060-uniform-partial-fresh-build.json`. This verifies the displayed uniform partial theorem, while literature priority, author approval and the full zero-constant target remain separate.

Publication strategy and attribution

Powerful-core decomposition, squarefree injectivity strategy and cutoff counting are prior machinery. The candidate original contribution is the exact weighted pattern improvement and its all-target entropy coefficient. The closest Kominers and related sources should be compared on their domains, constants and quantitative effectiveness before announcing best-known status.

A standalone analytic/computational number-theory note should supply the local rational inequality, the core injection and a transparent cutoff argument. The related cube-sum paper reuses some of this formal infrastructure, which should be acknowledged rather than counted as an independent discovery of the same lemmas.

This short manuscript records the construction and selected endpoints. The companion Original results anthology contains the extended ordinary proof exposition and its explicitly stated premises; the preserved Lean source remains the complete formal proof record.

Verification and reproducibility

Fresh execution: `sakana-erdos1060-uniform-partial`: PASS; 10/10 custom modules compiled successfully; receipt `Verification/sakana-erdos1060-uniform-partial-fresh-build.json`. Compiler pins, source hashes and endpoint axiom outputs are in `Verification`. Historical receipts and finite checks do not replace fresh replay.

Erdos1060.canonical_count_eventually_upper

```
theorem canonical_count_eventually_upper {ε : ℝ} (hε : 0 < ε) :
  ∀f n : ℕ in atTop,
  ({k ≤ n | k * ArithmeticFunction.sigma 1 k = n} : ℝ) ≤
  Real.exp (((Real.log (2539 / 1000) / 3 + Real.log (10 / 9) / 2) + ε) *
  Real.log (n : ℝ) / Real.log (Real.log (n : ℝ)))
```

Source: proofs/erdos1060-uniform-partial/src/SigmaFiberAsymptotic.lean:38.

Source SHA-256: 440f2f6aaa1d5dd04611fe855077352a3b75171d6e1ec823939ede8609e2d45a

Erdos1060.canonical_count_upper

```
theorem canonical_count_upper {ε : ℝ} (hε : 0 < ε) :
  ∃ N : ℕ, ∀ n : ℕ, N ≤ n →
  ({k ≤ n | k * ArithmeticFunction.sigma 1 k = n} : ℝ) ≤
  Real.exp (((Real.log (2539 / 1000) / 3 + Real.log (10 / 9) / 2) + ε) *
  Real.log (n : ℝ) / Real.log (Real.log (n : ℝ)))
```

Source: proofs/erdos1060-uniform-partial/src/SigmaFiberAsymptotic.lean:49.

Source SHA-256: 440f2f6aaa1d5dd04611fe855077352a3b75171d6e1ec823939ede8609e2d45a

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